Blob detection with OpenCL

Parallel programming project

Gutnar Leede & Grete-Lilijane Küppas

January 17, 2017

Introduction



Figure 1: Robotex football field

Tresholding



Figure 2: Source image

Tresholding

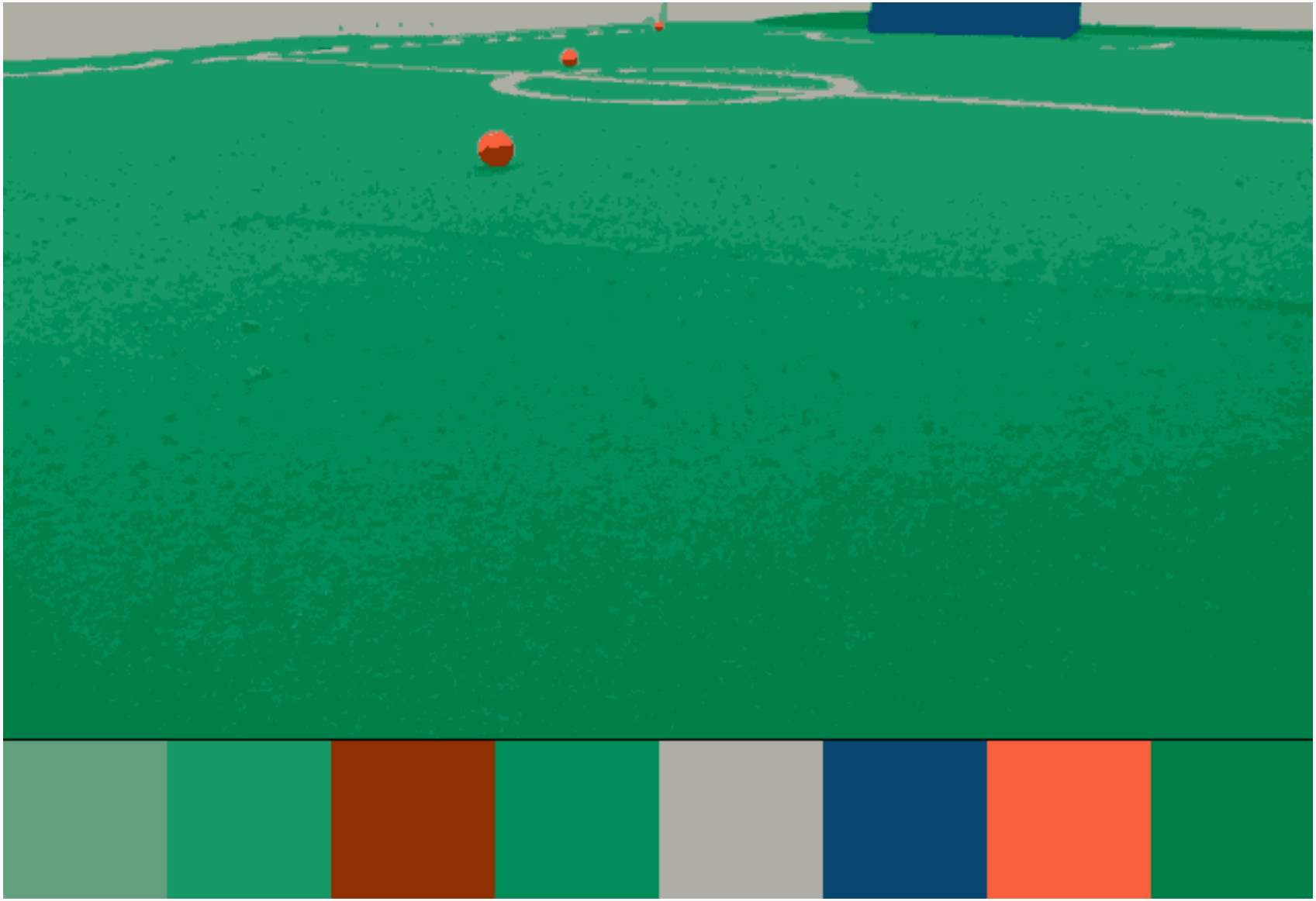


Figure 3: Tresholding result

Tresholding

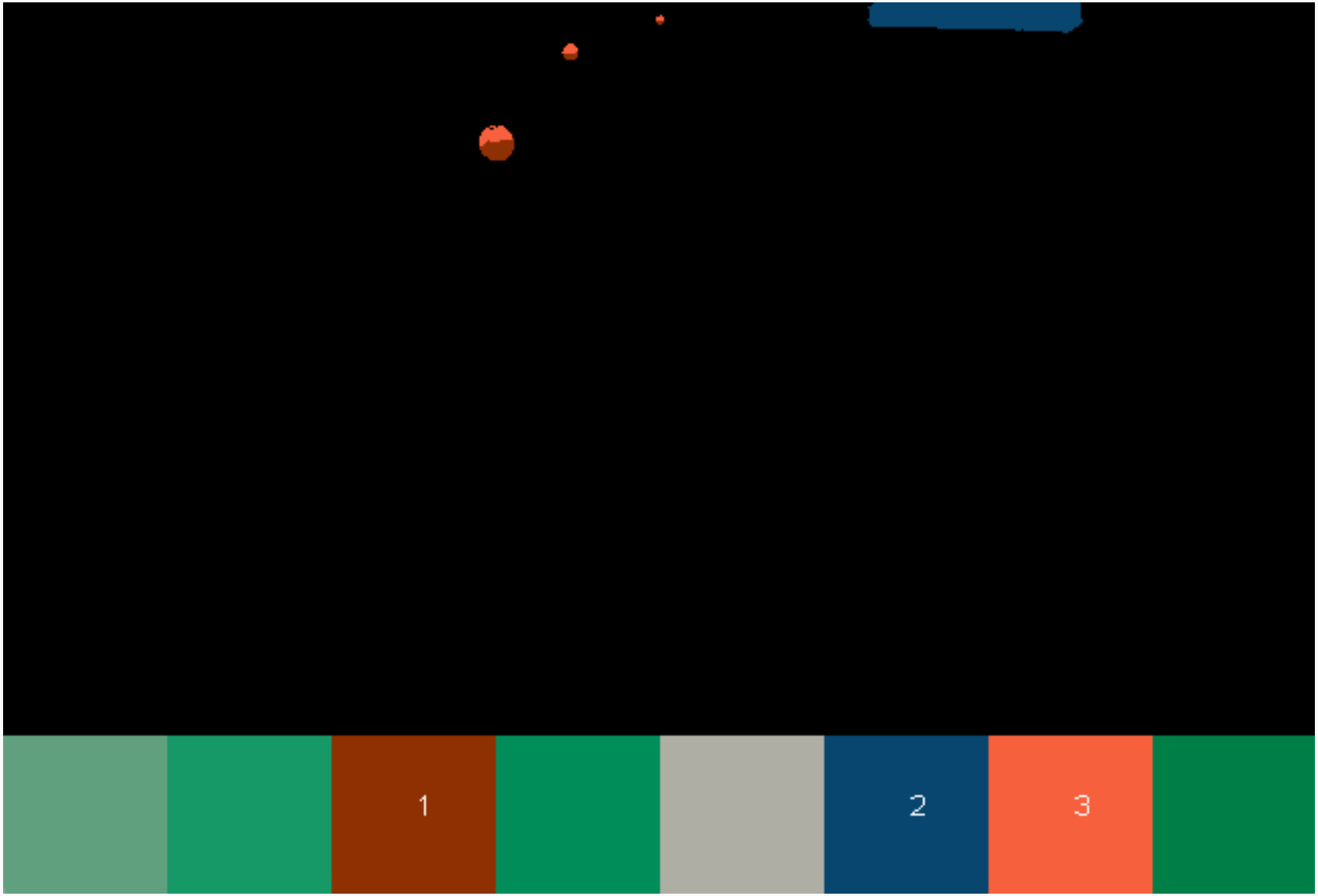


Figure 4: Selecting colors of interest

Blob detection

- Pixels are assigned into clusters
- Sequential same-colored pixels on a row are connected into chains
- Connected chains on adjacent rows are turned into blobs

Blob detection

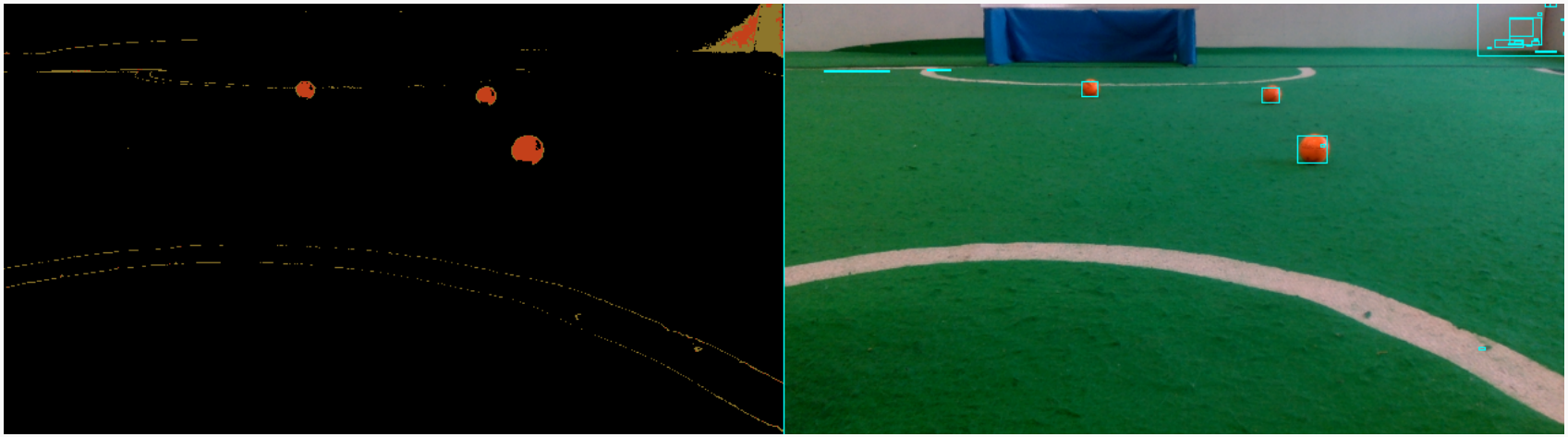


Figure 5: Detecting balls

Discerning similar colors



Figure 6: Objects with similar colors

Discerning similar colors

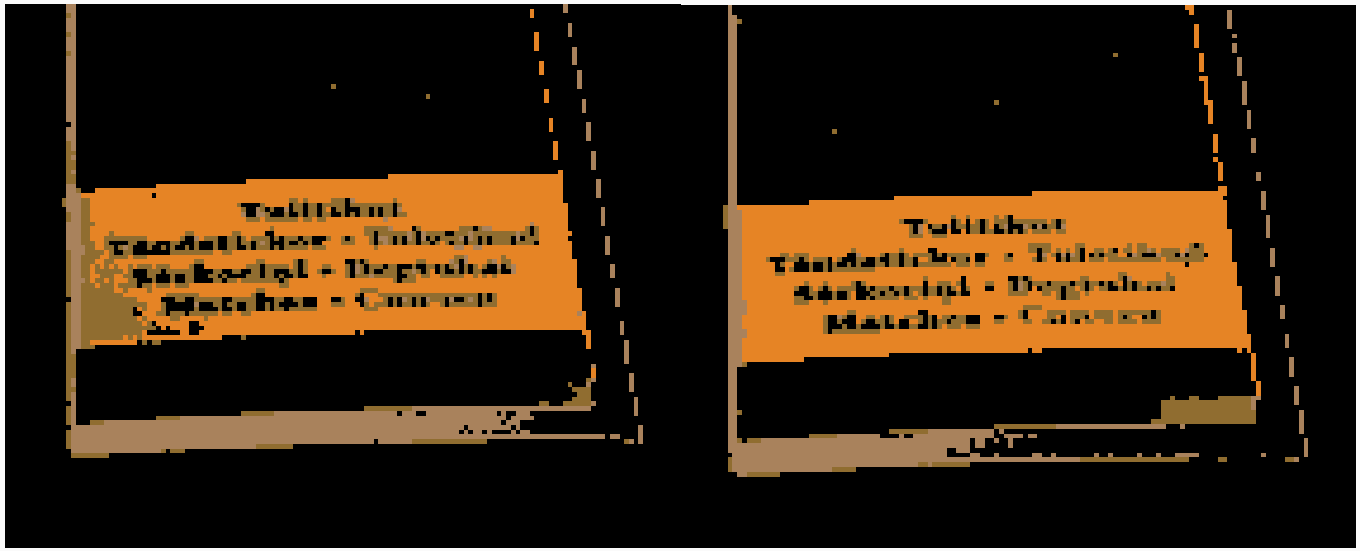


Figure 7: The blob on the right is more pure

Summary

- Learned to use OpenCL to perform parallel image processing
- Good starting point for Robotex 2017